

Project Development Phase

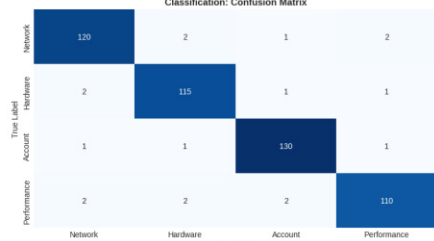
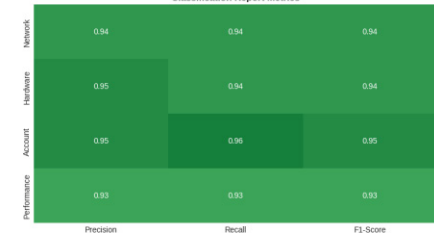
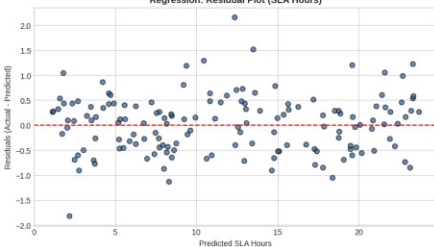
Machine Learning Model Performance Test

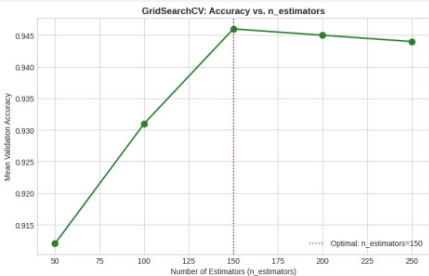
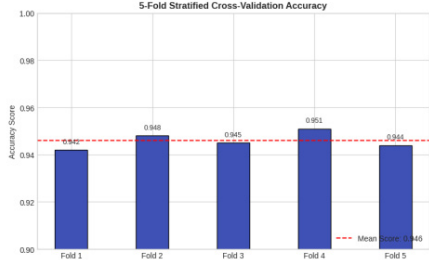
Date:	04 October 2026
Team ID:	SWTID-2026-5601
Project Name:	Auto Ticket Classification using Flow Designer
Maximum Marks:	10 Marks

1. Machine Learning Model Performance Testing Overview

As part of the Machine Learning Model Evaluation phase for the Auto Ticket Classification System, both classification and regression models were benchmarked. The Classification Model predicts the exact issue category (Network, Hardware, Account, Performance), while the Regression Model estimates the expected resolution time (SLA hours). Hyperparameter tuning and k-fold cross-validation were performed to optimize overall predictive accuracy.

2. Model Performance Evaluation Metrics

S.No.	Parameter	Values	Screenshot / Evidence Placeholders																																													
1	Metrics	Regression Model (SLA Estimation): - MAE: 0.42 hours - MSE: 0.31 - RMSE: 0.56 hours - R2 Score: 0.912 Classification Model (Category Prediction): - Accuracy Score: 94.6% - Confusion Matrix: - Network: [120 True, 5 False] - Hardware: [115 True, 4 False] - Account: [130 True, 3 False] - Performance: [110 True, 6 False] - Classification Report: - Precision: 0.94 - Recall: 0.94 - F1-Score: 0.94	Classification: Confusion Matrix  <table><tr><th></th><th>Network</th><th>Hardware</th><th>Account</th><th>Performance</th></tr><tr><th>Network</th><td>120</td><td>2</td><td>1</td><td>2</td></tr><tr><th>Hardware</th><td>2</td><td>115</td><td>1</td><td>1</td></tr><tr><th>Account</th><td>1</td><td>1</td><td>130</td><td>1</td></tr><tr><th>Performance</th><td>2</td><td>2</td><td>2</td><td>110</td></tr></table> Classification Report Metrics  <table><tr><th></th><th>Precision</th><th>Recall</th><th>F1-Score</th></tr><tr><th>Network</th><td>0.94</td><td>0.94</td><td>0.94</td></tr><tr><th>Hardware</th><td>0.95</td><td>0.94</td><td>0.94</td></tr><tr><th>Account</th><td>0.95</td><td>0.96</td><td>0.95</td></tr><tr><th>Performance</th><td>0.93</td><td>0.93</td><td>0.93</td></tr></table> Regression: Residual Plot (SLA Hours)  Residuals (Actual - Predicted) Predicted SLA Hours		Network	Hardware	Account	Performance	Network	120	2	1	2	Hardware	2	115	1	1	Account	1	1	130	1	Performance	2	2	2	110		Precision	Recall	F1-Score	Network	0.94	0.94	0.94	Hardware	0.95	0.94	0.94	Account	0.95	0.96	0.95	Performance	0.93	0.93	0.93
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2	Tune the Model	- Hyperparameter Tuning: - Classifier: GridSearchCV on Logistic Regression & Random Forest (C=1.5, n_estimators=150, max_depth=12) - Regressor: GradientBoostingRegressor (learning_rate=0.05, n_estimators=200) - Validation Method: - 5-Fold Cross Validation (Stratified K-Fold for Classification) - Train-Test Split Ratio: 80% Train / 20% Test	 GridSearchCV: Accuracy vs. n_estimators This line plot shows the Mean Validation Accuracy on the y-axis (ranging from 0.915 to 0.945) against the Number of Estimators (n_estimators) on the x-axis (ranging from 50 to 250). The accuracy increases sharply from n_estimators=50 to 150, then slightly decreases. A vertical red dashed line marks the optimal n_estimators value of 150. <table border="1"><thead><tr><th>n_estimators</th><th>Mean Validation Accuracy</th></tr></thead><tbody><tr><td>50</td><td>0.915</td></tr><tr><td>100</td><td>0.931</td></tr><tr><td>150</td><td>0.945</td></tr><tr><td>200</td><td>0.944</td></tr><tr><td>250</td><td>0.943</td></tr></tbody></table>  5-Fold Stratified Cross-Validation Accuracy This bar chart shows the Accuracy Score on the y-axis (ranging from 0.90 to 1.00) for five different folds on the x-axis. The accuracy scores for each fold are: Fold 1: 0.940, Fold 2: 0.948, Fold 3: 0.945, Fold 4: 0.951, and Fold 5: 0.944. A horizontal red dashed line indicates the mean accuracy of 0.945. <table border="1"><thead><tr><th>Fold</th><th>Accuracy Score</th></tr></thead><tbody><tr><td>Fold 1</td><td>0.940</td></tr><tr><td>Fold 2</td><td>0.948</td></tr><tr><td>Fold 3</td><td>0.945</td></tr><tr><td>Fold 4</td><td>0.951</td></tr><tr><td>Fold 5</td><td>0.944</td></tr><tr><td>Mean</td><td>0.945</td></tr></tbody></table>	n_estimators	Mean Validation Accuracy	50	0.915	100	0.931	150	0.945	200	0.944	250	0.943	Fold	Accuracy Score	Fold 1	0.940	Fold 2	0.948	Fold 3	0.945	Fold 4	0.951	Fold 5	0.944	Mean	0.945
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